

The empirical recurrence for OEIS A235898, proved by transfer matrix

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6 September 2026

Abstract

OEIS A235898 counts arrays of width 5 over $\{0, 1, 2\}$ in which the smallest and largest of the four entries of every 2×2 block have a sum equal to the sum of the other two. The entry carries an empirical recurrence of order 9 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical: the condition involves two consecutive rows and nothing wider, so the count is a walk count on $S = 243$ states.

2020 Mathematics Subject Classification. 05A15, 05B20, 15A18.

1 The conjecture

OEIS A235898 is “Number of $(n+1) \times (4+1)$ 0..2 arrays with the minimum plus the maximum equal to the lower median plus the upper median in every 2×2 subblock.”. It has offset 1 and begins

1797, 17001, 148945, 1384257, 12931393, 124394337, 1221438401, 12293017761, ...

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: $a(n) = 33*a(n-1) - 312*a(n-2) - 360*a(n-3) + 16976*a(n-4) - 22128*a(n-5) - 360576*a(n-6) + 253440*a(n-7) + 3520512*a(n-8) + 3870720*a(n-9)$

As of the “Last modified” line on the live entry (Jul 23 2025 08:54:47, revision 6) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

The arrays have 5 columns and a growing number of rows, with entries in $\{0, 1, 2\}$. Take any 2×2 block of the array and sort its four entries as

$$\alpha \leq \beta \leq \gamma \leq \delta.$$

The minimum is α and the maximum is δ ; the lower median is β and the upper median is γ . The entry’s condition, imposed on every 2×2 block, is therefore

$$\alpha + \delta = \beta + \gamma.$$

Nothing here depends on where in the block each value sits: the condition is a statement about the multiset of the four entries. In particular it is symmetric under every rearrangement of the block, and under adding a constant to all four entries.

3 Two rows

A 2×2 block occupies two consecutive rows, so everything the condition asks is decided by a pair of consecutive rows and nothing wider. Take the array one row at a time and let the state be the row just written; a step appends a new row and is allowed exactly when every one of the $5 - 1$ blocks spanning the two rows satisfies the condition.

Every block of the finished array is tested exactly once, at the step that writes its lower row, and no step tests anything the definition does not ask. A one-row array has no block at all, so every row may start and every state may end: ι and τ are both $\mathbf{1}$, and the states are all $3^5 = 243$ rows. Thus

$$a(n) = \iota^\top M^j \tau, \quad S = 243,$$

for the transfer matrix M on those S states. The walk index j and the entry's own n differ by a constant, read off by matching the published terms rather than assumed. In particular a is C -finite.

4 The proof

Lemma 1. *Let $q(t) = t^9 - \sum_i c_i t^{9-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M) \tau$ for the corresponding walk index j , so the recurrence holds from some point on if and only if $\iota^\top M^j q(M) \tau = 0$ for all large j , and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+9} - \sum_i c_i M^{j+9-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M) \tau$ obey the monic recurrence given by the characteristic polynomial of M ; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 2.

$$a(n) = 33a(n-1) - 312a(n-2) - 360a(n-3) + 16976a(n-4) - 22128a(n-5) - 360576a(n-6) + 253440a(n-7) + 360576a(n-8) - 33a(n-9)$$

for every $n > 8$.

Proof. The vector $w = q(M)\tau$ was formed by 9 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \dots$, again exactly; those integers vanish from the index corresponding to $n = 8 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. $\square$

5 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic.

Second, the count was recomputed for the smallest arrays straight from the definition: every array was written out cell by cell and each 2×2 block tested by sorting its four values and comparing the outer two with the inner two. No rows, no state and no transfer matrix are involved in that computation, and the published terms came back.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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- [3] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.